CELESTIAL MECHANICS × AI · 2026

# From the next token to the next theorem

How a model works — and how I use it in practice



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# What the model sees: ten integers

EN · ONE SENTENCE

Celestial mechanics is the poetry of orbits.

| Cel | estial | _mechanics | _is | _the | _poetry | _of | _or | bits | . |
|---|---|---|---|---|---|---|---|---|---|
| 69126 | 90010 | 52347 | 382 | 290 | 38153 | 328 | 503 | 24043 | 13 |

**46** characters → **10** tokens → 10 integers · _ marks a leading space

CHINESE · FOUNDATIONS OF CELESTIAL MECHANICS

本书系统地介绍了天体力学

| 本 | 书 | 系统 | 地 | 介绍 | 了 | 天 | 体 | 力 | 学 |
|---|---|---|---|---|---|---|---|---|---|
| 4087 | 20121 | 31936 | 5557 | 37887 | 4531 | 867 | 7074 | 11343 | 8590 |

**12** characters → **10** tokens → 10 integers · two-character words often map to one token

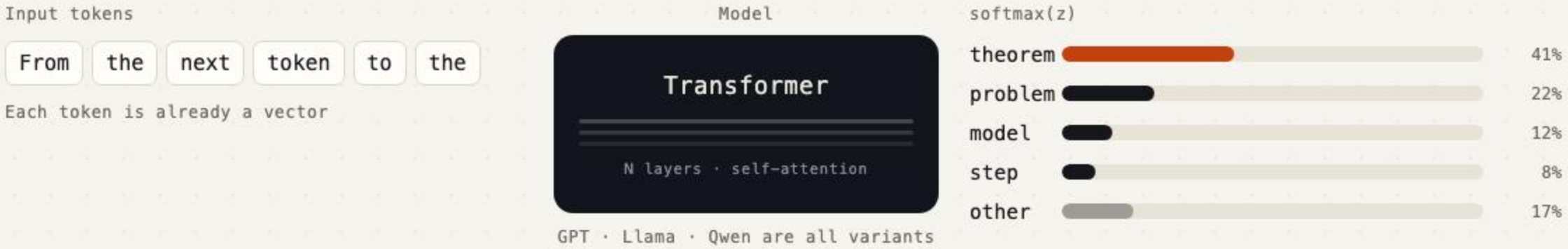
“This book systematically introduces celestial mechanics.”

01 Where do these integers come from?

02 How does it predict the next one?

03 How do we put it to work?

# What the model does: input → probability distribution




$$z = f_\theta(x_{1:t}) \in \mathbb{R}^{|V|} \qquad p_\theta(y \mid x_{1:t}) = \frac{\exp(z_y)}{\sum_{v \in V} \exp(z_v)}$$

$|V| = 200{,}019$

$\Sigma_y\, p_\theta(y \mid x_{1:t}) = 1$

> A scorer: it gives every token in the vocabulary a **score**, then turns scores into **probabilities**.

# RL: actions, outcomes and feedback

Act in an environment; use outcome feedback to improve action choices.

Observe environment → Take action → Outcome → Feedback / reward → Improve policy ↺

ENVIRONMENT

### Produce an outcome

Actions affect the environment; new observations return.

FEEDBACK

### Evaluate the result

Reward expresses progress or success.

POLICY

### Improve action choices

Use experience to seek higher long-run return.

> RL uses **outcome feedback** to learn **how to choose actions**.

Reference: Sutton & Barto, Reinforcement Learning: An Introduction.

# A score can teach a strategy

Breakout: move the paddle, hit the ball, clear the bricks.

TRY

### Different moves

Explore timing and direction.

OBSERVE

### Score the outcome

Miss the ball, or clear bricks.

LEARN

### Better strategies

Learn actions with delayed benefits.

> Repeated trials + outcome feedback → a policy that seeks higher reward.

> A useful environment lets the agent **try, evaluate and try again**.

Example: Mnih et al., Nature (2015), Breakout demonstration.

# Math and code share a feedback advantage

01 · CLOSED ENVIRONMENT

### Axioms · Runtime

Math uses definitions, axioms and inference rules; code uses a language spec and runtime. The system is self-contained.

02 · TRACEABLE ERRORS

### Locate the step

Proofs check inference step by step; tests, traces and line numbers locate code errors.

03 · CLEAN DATA

### Parse and filter

Formal objects and code are structured, enabling bulk generation and verification.

Two levers: better policies + more compute for candidate generation and verification.

Closed environment + localizable errors + clean data → **cheap feedback, large-scale trial and error**.

Closure is relative: code still depends on inputs and libraries; modeling still needs formalization; data still needs curation.

# Summary: environment, feedback, updates

Well-defined tasks let a model predict, receive checks and iterate in an environment.

Problem + environment → Next-token prediction → Check + trace → Define loss → Backpropagate ↺

01 · ENVIRONMENT

## Clear rules & goal

The model can act, try and produce outputs.

02 · FEEDBACK

## Evaluate & trace

Check outcomes and locate failing steps.

03 · UPDATE

## Loss-driven iteration

Backpropagate, update parameters, improve.

> The environment supplies feedback; **loss turns it into parameter updates**.

# When code outruns human review

Claude Code's review problem: code scales faster than human attention.

SCALE PRESSURE · ANTHROPIC MONOREPO

**2,500 → 30,000+**

Changes landing per day · May → Sep 2026

Claude Code has a separate repository.

PRODUCTION

Agents write more code, faster.

BOTTLENECK

Human review time stays limited.

CONSEQUENCE

Line-by-line review cannot keep up.

Goal: reduce **review + repair time**, with reverts and incidents as guardrails.

The next gain comes from **redesigning review**.

Source: Osmani, Lunt & Wang · 15 Oct 2026 · How the Claude Code team does code review · Team-reported practice.

# Acceleration: from basic use to gold medals

2023 · BEFORE

## Basic use

- Explain concepts, search the literature, write code
- Do simple calculations, often wrong
- Complex proofs still need people
- Role: assistant

→ 2024–26

2025–26 · AFTER

## Research level

- IMO 2025 gold level 35/42
- Lean proofs, machine-checkable
- OpenAI 2026: ≈3 Pro-hours of compute per result*
- Role: candidate generator

2024.07 **28/42** silver → 2025.07 **35/42** gold level 12 months, +7 points

* Compute equivalent, not elapsed time · OpenAI, 6 Oct 2026 · Company report.

> This is a staircase with **three steps**.

# From the next token to the next theorem

`token` → `probability` → `candidate` → `verification` → **`theorem`**

Probability gives breadth

Verification gives correctness

**People set the direction**

> The next theorem still begins with the **next token**.